

Data Preparation Notes

Customer Segmentation Project

1. Missing Values

- Profession → Replaced nulls with "Unknown" to retain records without discarding customers.
- Work_Experience → Imputed missing values with the median of the column to avoid skew from outliers.
- Var_1 → Filled nulls with "Cat_0" as a placeholder category.
- Other categorical fields (Gender, Ever_Married, Graduated, Spending_Score) → Missing entries replaced with "Unknown" or "Not Specified".

2. Duplicates

- Checked for duplicate rows using customer ID.
- Removed exact duplicates to ensure each customer record is unique.
- Verified that no duplicate IDs remained in Train or Test datasets.

3. Inconsistencies

- Standardized categorical values:
 - Gender → {Male, Female, Unknown}
 - Spending_Score → {Low, Average, High}
- Normalized inconsistent text cases (e.g., "yes" vs "Yes").
- Family_Size → Converted missing values to median family size.
- Age → Verified ranges and corrected unrealistic ages.

4. Categorical Encoding

- Applied Label Encoding for categorical variables (Gender, Profession, Var_1, Spending_Score).
- Ensured Train and Test datasets used the same encoding mappings.

5. Scaling

- Standardized numeric features (Age, Work_Experience, Family_Size) using StandardScaler.
- Ensured Train and Test datasets were scaled consistently.

6. Final Outputs

- Train_cleaned.csv → Cleaned training dataset with no missing values, duplicates, or inconsistencies.
- Test_cleaned.csv → Cleaned test dataset with identical preprocessing rules.
- Train_ready.csv / Test_ready.csv → Encoded and scaled datasets ready for segmentation modeling.